Lab 13. Working with Keys and Vault to Perform Cryptographic Operations

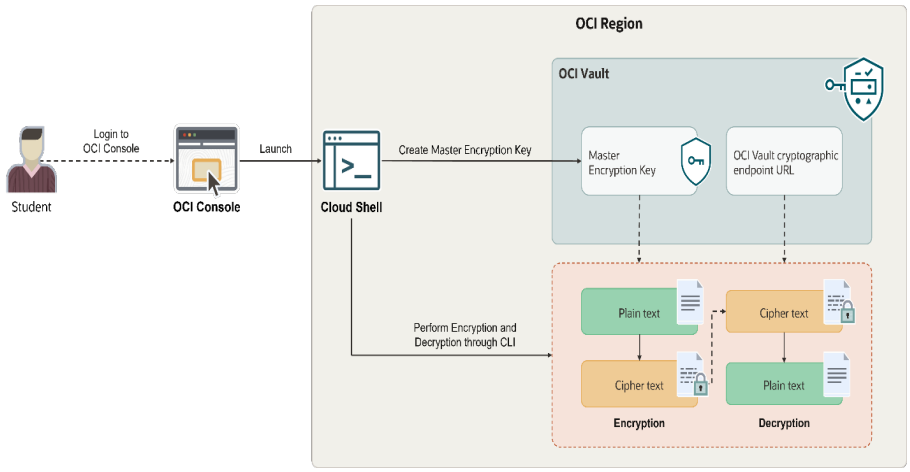
Overview

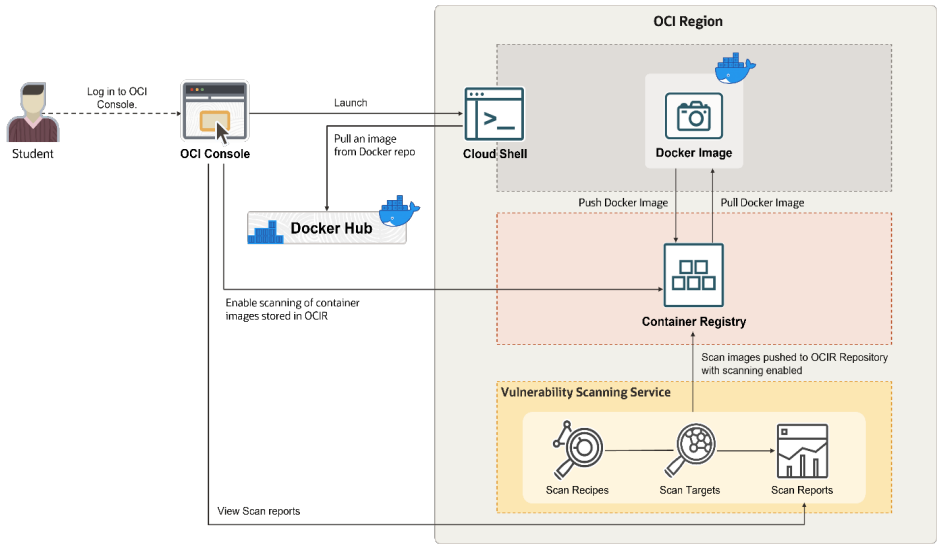
Oracle Cloud Infrastructure (OCI) Vault is a key management service that stores and manages master encryption keys and secrets for protected resource access. Depending on the protection mode, vault keys are either saved on the server or stored in highly accessible and robust hardware security modules (HSM).

Critical to shipping compliant and secure containers, system administrators need to ensure that container images have the following characteristics:

- Are free of known critical vulnerabilities that can cause an accidental system failure or result in malicious activity.
- Have not been modified since they were published to maintain their integrity.
- Are only deployed to a Kubernetes cluster and come from a trusted source.

OCI container image scanning, signing, and verification address all these secure container deployment needs.





In this lab, you'll:

- Prepare for master encryption key
- Create master encryption key
- Prepare for encryption and decryption
- Perform encryption and Perform decryption
- Rotate the master encryption key
- Create an auth token
- Create a new container repository
- Enable image scanning
- Sign in to OCIR from the Cloud Shell
- Pull the Docker image from Docker hub
- Tag the Docker image
- Push the tagged Docker image to OCIR
- Verify if the image has been pushed
- View scan results
- View vulnerability reports

Sequence 1. Encryption-Decryption with Vault and Keys

Prepare for Master Encryption Key

To create a master encryption key, you'll need the compartment OCID and management endpoint URL.

While you're performing these tasks, copy and save the compartment OCID, Vault OCID, and management endpoint URL in a notepad.

1. Launch the Jupyter Lab server and switch to root user and get the compartment OCID.

   ```
   $ sudo su -
   ```

   ```
   # oci iam compartment list --name <assigned_compartment>| grep '<assigned_compartment>\|ocid1.compartment'
   ```

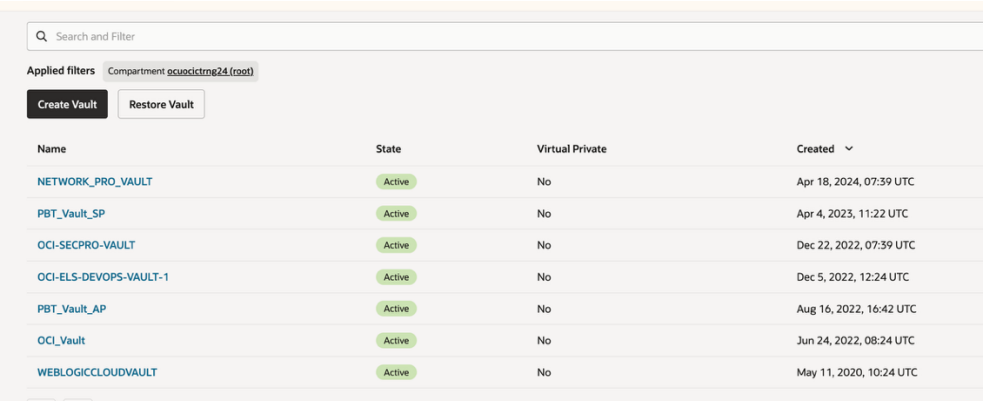
   Replace the *<assigned_compartment>* with the compartment name that is assigned to you. For example,

   ```
   # oci iam compartment list --name 98631166-C16| grep '98631166-C16\|ocid1.compartment'
   ```

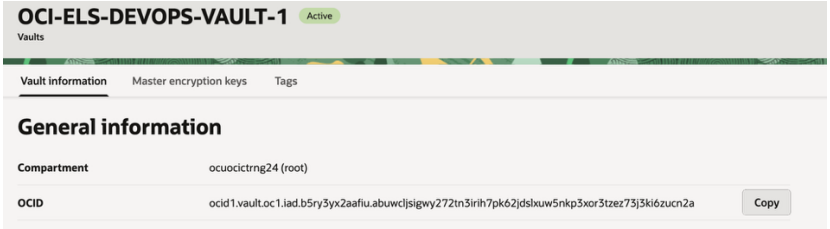
   Sample Output:

2. Get the management endpoint URL.
   a. You will need a Vault OCID to get the management endpoint URL. To get the Vault OCID:
      1) Open the navigation menu, click **Identity & Security**, and then click **Vault**.
      2) Select ***root*** compartment from ***Applied filters.***

3) From the list of vaults, click the vault **OCI-ELS-DEVOPS-VAULT-1**.



4) Copy the Vault OCID and paste it to a notepad for later use. OCID is the unique Oracle-assigned ID of the vault.



b. Generate and copy the management endpoint URL in a notepad:

```
oci kms management vault get --vault-id <vault_ocid> | grep
'display-name\|management-endpoint'
```

Replace *<vault_ocid>* with the vault OCID you copied earlier. For example,

```
oci kms management vault get --vault-id
ocid1.vault.oc1.iad.b5ry3yx2aafiu.abuwcljsigwy272tn3irih7pk62jdslxuw5nkp3xor
3tzez73j3ki6zucn2a | grep 'display-name\|management-endpoint'
```

Sample Output: Copy the URL highlighted below in bold.

```
[root@aiops-labs ~]# oci kms management vault get --vault-id ocid1.vault.oc1.iad.b5ry3yx2aafiu.abuwcljsigwy272tn3irih7pk62jdslxuw5nkp3xor3tzez73j3ki6zucn2a |
grep 'display-name\|management-endpoint'
    "display-name": "OCI-ELS-DEVOPS-VAULT-1",
    "management-endpoint": "https://b5ry3yx2aafiu-management.kms.us-ashburn-1.oraclecloud.com",
[root@aiops-labs ~]#
```

Create Master Encryption Key

You will create a master encryption key needed to carry out cryptographic operations using an existing Vault at the root level compartment.

1. Create a master encryption key.

```
oci kms management key create \
--compartment-id <compartment_ocid> \
--display-name <key_name> \
--key-shape <key_encryption_information> \
```

```
--endpoint <management_endpoint_url>
```

Replace *<compartment_ocid>* and *<management_endpoint_url>* with the compartment OCID and the management endpoint URL you saved earlier.

Replace *<key_name>* with IAD-DP-LAB19-1-MSK-01 and *<key_encryption_information>* with '{"algorithm":"AES","length":"16"}'. For example,

```
oci kms management key create \
--compartment-id ocid1.compartment.oc1..aaaaaaaa7mt4nzrixxxxxxxxxxxx\
--display-name IAD-DP-LAB19-1-MSK-01 \
--key-shape '{"algorithm":"AES","length":"16"}' \
--endpoint https://b5ry3yx2aafiu-management.kms.us-ashburn-1.oraclecloud.com
```

Sample Output:

```
[root@aiops-labs ~]# oci kms management key create \
--compartment-id ocid1.compartment.oc1..aaaaaaaawjohov7fktaf7b7hhqkh7xtjjn2l44pnwcy42dhvqjziyllkxokq \
--display-name IAD-DP-LAB19-1-MSK-01 \
--key-shape '{"algorithm":"AES","length":"16"}' \
--endpoint https://b5ry3yx2aafiu-management.kms.us-ashburn-1.oraclecloud.com
{
  "data": {
    "auto-key-rotation-details": null,
    "compartment-id": "ocid1.compartment.oc1..aaaaaaaawjohov7fktaf7b7hhqkh7xtjjn2l44pnwcy42dhvqjziyllkxokq",
    "current-key-version": "ocid1.keyversion.oc1.iad.b5ry3yx2aafiu.aw4cnhpywaiaa.abuwcljtdkml6i2jzxpyq76ijohsr6tnt5g33qp4zwxd3j3dbcqtdvf23mxa",
    "defined-tags": {},
    "display-name": "IAD-DP-LAB19-1-MSK-01",
    "external-key-reference-details": null,
    "freeform-tags": {},
    "id": "ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq",
    "is-auto-rotation-enabled": false,
    "is-primary": true,
    "key-shape": {
      "algorithm": "AES",
      "curve-id": null,
      "length": 16
    },
    "lifecycle-state": "CREATING",
    "protection-mode": "HSM",
    "replica-details": null,
    "restored-from-key-id": null,
    "time-created": "2026-04-15T12:56:33.784000+00:00",
    "time-of-deletion": null,
    "vault-id": "ocid1.vault.oc1.iad.b5ry3yx2aafiu.abuwcljsigwy272tn3irih7pk62jdslxuw5nkp3xor3tzez73j3ki6zucn2a"
  },
  "etag": "8721243e2e27509aa4bf1a1d19349a5790cfb583--gzip"
}
[root@aiops-labs ~]#
```

Prepare for Encryption and Decryption

To perform encryption and decryption, you will need the master encryption key OCID and OCI Vault cryptographic endpoint URL to perform encryption.

Copy and save master encryption key OCID (ID in the output) and OCI Vault cryptographic endpoint URL in a notepad. You'll be using these OCIDs and URL later in this lab.

1. Get the master encryption key OCID and copy in notepad.

```
oci kms management key list \
--compartment-id <compartment_ocid> \
--endpoint <management_endpoint_url> | \
grep 'display-name\|ocid1.key'
```

Replace *<compartment_ocid>* and *<management_endpoint_url>* with the compartment OCID and the management endpoint URL you saved earlier. For example,

```
oci kms management key list \
--compartment-id
ocid1.compartment.oc1..aaaaaaaawjohov7fktaf7b7hhqkh7xtjjn2l44pnwcy42
dhvqjziyllkxokq \
--endpoint https://b5ry3yx2aafiu-management.kms.us-ashburn-
1.oraclecloud.com \
```

```
 | grep 'display-name\|ocid1.key'
```

Sample Output:

```
[root@aiops-labs ~]# oci kms management key list \
--compartment-id ocid1.compartment.oc1..aaaaaaaawjohov7fktaf7b7hhqkh7xtjjn2l44pnwcy42dhvqjziyllkxokq \
--endpoint https://b5ry3yx2aafiu-management.kms.us-ashburn-1.oraclecloud.com \
 | grep 'display-name\|ocid1.key'
WARNING: This operation supports pagination and not all resources were returned.  Re-run using the --all option to auto paginate and list all resources.
      "display-name": "IAD-DP-LAB19-1-MSK-01",
      "id": "ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq",
      "display-name": "iad-dop-lab05-1-vk-01",
      "id": "ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtsa7pgq7jd5vubsbs7sghju65gws72gmg36kx7zhn7rypwlaupm7a",
[root@aiops-labs ~]#
```

2. Get the cryptographic endpoint URL and copy in notepad.

```
oci kms management vault get \

--vault-id <vault_ocid> | grep 'display-name\|crypto-endpoint'
```

Replace *<vault_ocid>* with the Vault OCID you saved earlier. For example,

```
oci kms management vault get --vault-id
ocid1.vault.oc1.iad.bzqtr2wtaacuu.abuwcljrfxxxxxxxxxxxxxxxt5aq |
grep 'display-name\|crypto-endpoint'
```

Sample Output:

```
[root@aiops-labs ~]# oci kms management vault get --vault-id ocid1.vault.oc1.iad.b5ry3yx2aafiu.abuwcljsigwy272tn3irih7pk62j
dslxuw5nkp3xor3tzez73j3ki6zucn2a | grep 'display-name\|crypto-endpoint'
    "crypto-endpoint": "https://b5ry3yx2aafiu-crypto.kms.us-ashburn-1.oraclecloud.com",
    "display-name": "OCI-ELS-DEVOPS-VAULT-1",
[root@aiops-labs ~]#
```

Perform Encryption

You will execute CLI commands to perform encryption. The CLI command invokes `oci kms crypto encrypt` to perform data encryption which will require three inputs: the plain text to be encrypted, the OCID of the master encryption key you created in the last step, and the OCI Vault cryptographic endpoint.

1. Perform encryption.

```
$ oci kms crypto encrypt \
--key-id <master_encryption_key_OCID> \
--endpoint <crypto_endpoint> \
--plaintext "base64_plain_text"
```

Replace `<master_encryption_key_OCID>` and `<crypto_endpoint>` with the master key OCID and crypto endpoint you saved in the previous task. Additionally, add a base64 plain text for encryption. For example,

```
oci kms crypto encrypt \
--key-id ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljzf4unp3vy6f6ka \
--endpoint https://b5ry3yx2aafiu-crypto.kms.us-ashburn-1.oraclecloud.com \
--plaintext "$(echo HELLO_WORLD | base64)"
```

Sample Output:

```
[root@aiops-labs ~]# oci kms crypto encrypt \
--key-id ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq \
--endpoint https://b5ry3yx2aafiu-crypto.kms.us-ashburn-1.oraclecloud.com \
--plaintext "$(echo HELLO_WORLD | base64)"
{
  "data": {
    "ciphertext": "QQbMk2l7dBI7AHyIk3YIHJ+zeDeILYmiVmnIst2BVI2ZCF/9YwIw+pRZSu+jWNcp7W0AAAAA",
    "encryption-algorithm": "AES_256_GCM",
    "key-id": "ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq",
    "key-version-id": "ocid1.keyversion.oc1.iad.b5ry3yx2aafiu.aw4cnhpywaiaa.abuwcljtdkml6i2jzxpyq76ijohsr6tnt5g33qp4zwxd3j3
dbcqtdvf23mxa"
  }
}
```

2. Copy **ciphertext** from your output as highlighted using arrow in the sample output and paste it in a notepad to use later during the decryption process.

Perform Decryption

You will execute CLI commands to perform decryption. The CLI command invokes `oci kms crypto decrypt` to perform data encryption which will require three inputs: the encrypted plain text that needs to be decrypted, the OCID of the master encryption key you created in the previous step, and the OCI Vault cryptographic endpoint.

1. Perform decryption.

```
$ oci kms crypto decrypt \
--key-id <master_encryption_key_OCID> \
--endpoint <crypto_endpoint> \
--ciphertext <ciphertext>
```

Replace the *<master_encryption_key_OCID>*, *<crypto_endpoint>*, and the *<ciphertext>* with the master key OCID, crypto endpoint, and the ciphertext respectively you saved earlier. For example,

```
oci kms crypto decrypt \
--key-id ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq \
--endpoint https://b5ry3yx2aafiu-crypto.kms.us-ashburn-1.oraclecloud.com \
--ciphertext QQbMk2l7dBI7AHyIk3YIHJ+zeDeILYmiVmnIst2BVI2ZCF/9YwIw+pRZSu+jWNcp7W0AAAAA
```

Sample Output:

```
[root@aiops-labs ~]# oci kms crypto decrypt \
--key-id ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq \
--endpoint https://b5ry3yx2aafiu-crypto.kms.us-ashburn-1.oraclecloud.com \
--ciphertext QQbMk2l7dBI7AHyIk3YIHJ+zeDeILYmiVmnIst2BVI2ZCF/9YwIw+pRZSu+jWNcp7W0AAAAA
{
  "data": {
    "encryption-algorithm": null,
    "key-id": "ocid1.key.oc1.iad.b5ry3yx2aafiu.abuwcljtkkjfogrciiaus2vxumvqwhos3bmovmh3vbfwoj763jqqoxqh6hxq",
    "key-version-id": "ocid1.keyversion.oc1.iad.b5ry3yx2aafiu.aw4cnhpywaiaa.abuwcljtdkml6i2jzxpyq76ijohsr6tnt5g33qp4zwxd3j3
dbcqtdvf23mxa",
    "plaintext": "SEVMTE9fV09STEQK",
    "plaintext-checksum": "3569994667"
  }
}
[root@aiops-labs ~]#
```

2. Copy the **plaintext** code from your output as highlighted using arrow in the sample output and paste it in a notepad to use it during decoding process.
3. Decode base64 plaintext:

```
$ echo <plaintext_code> | base64 --decode
```

```
[root@aiops-labs ~]# echo "SEVMTE9fV09STEQK"| base64 --decode
HELLO_WORLD
[root@aiops-labs ~]#
```

Replace the *<plaintext_code>* with the plaintext code you saved in the previous step.

Purge Instructions

Purge Instructions for Master Encryption Key

1. Sign in to Oracle Cloud Infrastructure.
2. Open the navigation menu, click **Identity & Security**, and then click **Vault**.
3. Select **root** compartment from List scope on the left menu.
4. From the list of vaults, click the vault **OCI-ELS-DEVOPS-VAULT-1.**
5. Select your *<assigned compartment>* from **Applied filters**.
6. Click **Master Encryption Keys** and locate the key with the name **IAD-DP-LAB13-1-MSK- 01**.
7. Click the three dots on the right to open the Actions menu. Select **Delete Key**.
   - Confirm that you want to delete the key by clicking the box and then typing the key name.
   - Schedule when you want the Vault service to delete the key. You can set a date after 8 days.
8. Click **Delete Key**.

Sequence 2. Image Signing and Scanning Create an Auth Token

Create an auth token to use when authenticating with your *<assigned user account>* with Oracle Cloud Infrastructure Registry (OCIR).

1. In the top-right corner of the OCI Console, open the **Profile** menu, and then click **User settings**.
2. Click on **Tokens and Keys** tab. Now under **Auth tokens** click **Generate Token**.

   A user can only have two auth tokens at a time. Delete an older auth token if you need to create a new one.

3. Enter **IAD-DOP-LAB13-1-AT-01**, as a friendly description for the auth token.

   Click **Generate Token**. The new auth token is displayed. Click on **Copy** from by clicking on three dots at the end of the auth token to copy and paste it in a notepad because you won't see the auth token again in the Console.

4. Click **Close**.

Create a New Container Repository

Create a repository in your assigned compartment and give it a name that's unique across all compartments in the entire tenancy. Having created the new repository, you can push an image to the repository using the Docker CLI.

1. Sign in to the Oracle Cloud Infrastructure (OCI) Console.
2. Open the navigation menu and click **Developer Services.** Under **Containers & Artifacts**, click **Container Registry**.
3. Select your *<assigned compartment>* from List scope on the left menu.
4. Click **Create repository**.
5. Select your *<assigned compartment>* to create a new repository.
6. Enter a name for the new repository:

   <region-key>-aiops-lab13-ocir-1/demo_imagescan_*<userID>*

For example, `iad-aiops-lab13-ocir-1/demo_imagescan_user16`

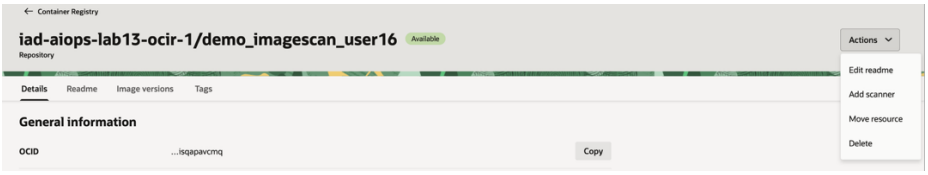
Only use lower case characters, numbers, and special characters.

7. Copy the `<repo-name>` on a notepad for use later in this lab.
8. Select the **Private** option to limit access to the new repository and Click **Create Repository**.

Enable Image Scanning

When you create a new repository, image scanning is disabled by default. You can use the Console to enable image scanning for a repository by creating a new image scanner. If image scanning has already been enabled, you can use the Console to disable it.

1. Once the repository is created, click the newly created repository `<region-key>-dop-lab20-1-ocir-1/oci_demo_imagescan_<userID>` from the list of repositories.
2. From **Actions** menu displayed on extreme right and select **Add scanner**.

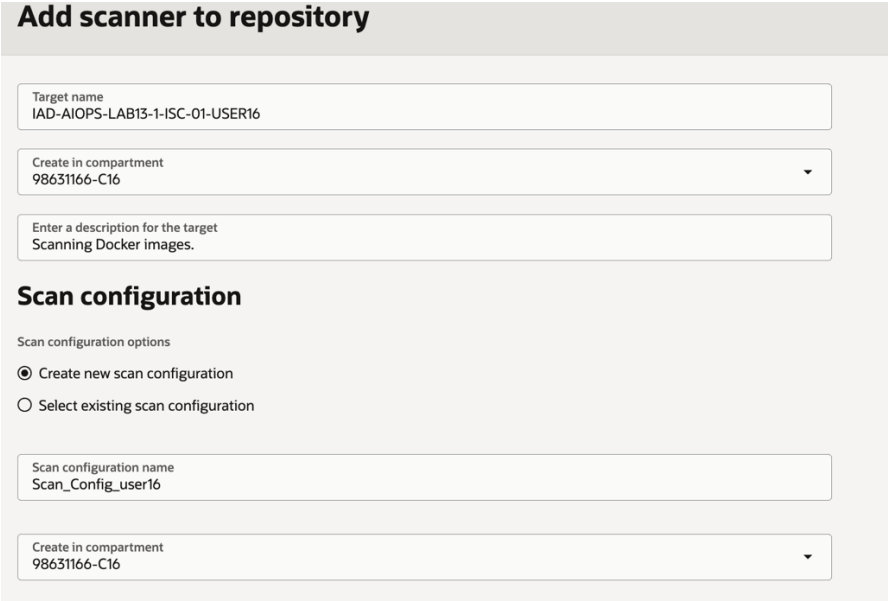


3. In the **Add scanner to repository** pane, enter the following values:
   - **Name:** `IAD-AIOPS-LAB13-1-ISC-01-<userID>`

     For example, `IAD-AIOPS-LAB13-1-ISC-01-USER16`

   - **Create in compartment:** `Select your <assigned compartment>.`
   - **Description (Optional):** `Scanning Docker images.`
   - Select **Create a new scan configuration**.
   - **Name**: `Scan_Config_<userID>`

     For example, `Scan_Config_user16`



4. Click **Add Scanner**.

   Now that a scanner has been created and configured, images saved to the repository will be scanned for vulnerabilities. If the repository already contains images, the four most recently saved images will have been immediately scanned for vulnerabilities when the scanner was created.

Sign In to OCIR from the Cloud Shell

Once you have generated the auth token and created a new repository, sign in to OCIR.

1. Switch to Jupyter Lab Server, Launch terminal and ensure that you are working as root user.
2. Log in to OCIR by entering:

   `podman login <region-key>.ocir.io`

   For example,

   `podman login iad.ocir.io`

   If is shows already logged in, logout with `podman logout iad.ocir.io`

3. When prompted, enter your username in the format given below.

   `<tenancy-namespace>/<username>`.

   Enter the auth token IAD-DOP-LAB13-1-AT-01 you copied earlier as the password.

```
[root@aiops-labs ~]# podman logout iad.ocir.io
Removed login credentials for iad.ocir.io
[root@aiops-labs ~]# podman login iad.ocir.io
Username: ocuocictrng24/98631166-lab.user16
Password:
Login Succeeded!
[root@aiops-labs ~]#
```

Pull the Docker Image

Begin by copying a prebuilt `maven` image tagged `latest` `(maven:latest)` from Docker Hub. This image will later be tagged and pushed to your OCIR repository.

1. Issue the following Docker pull command:

   ```
   podman pull maven:latest
   ```

   Select docker.io using down arrow key. And press enter. For example,

   ```
   [root@aiops-labs ~]# podman pull maven:latest
   ✔ docker.io/library/maven:latest
   Trying to pull docker.io/library/maven:latest...
   Getting image source signatures
   Copying blob 131bd6f641c5 done   |
   Copying blob d58dcb4ab265 done   |
   Copying blob 689b91d88a0f done   |
   Copying blob e1d7a89f363d done   |
   Copying blob 4f4fb700ef54 done   |
   Copying blob abc49b9bffc8 done   |
   Copying blob f3bce9374816 done   |
   Copying blob c19daefeb9d5 done   |
   Copying blob 4f4fb700ef54 skipped: already exists
   Copying blob a3e4261355ba done   |
   Copying config 87cfb96256 done   |
   Writing manifest to image destination
   87cfb9625636d97e4cda88b124314abee7f162e101ee8c2790fe460a1f74e1dc
   [root@aiops-labs ~]#
   ```

2. Verify the Docker pull command successfully executed:

   ```
   podman images
   ```

   You should see the `maven:latest` image in the list of images.

Tag the Docker Image

A tag identifies the Oracle Cloud Infrastructure Registry region, tenancy, and repository to which you want to push the image.

1. Run the following command to attach a tag to the image that you're going to push to OCIR repository:

   ```
   podman tag maven:latest <region-key>.ocir.io/<tenancy-namespace>/<repo-name>:<tag>
   ```

   For example,

   ```
   podman tag maven:latest iad.ocir.io/ocuocictrng24/iad-aiops-lab13-ocir-1/demo_imagescan_user16:1.0
   ```

2. Validate if the new image with the tag is listed.

   ```
   podman images
   ```

   Although two tagged images will be shown `(latest` and `1.0)`, both are based on the

same image with the same `IMAGE_ID`.

Push the Tagged Docker Image to OCIR

After assigning a tag to the image, push it to Oracle Cloud Infrastructure Registry repository.

1. Run the following command to push the tagged Docker image to OCIR repository:

   `podman push <region-key>.ocir.io/<tenancy-namespace>/<repo-name>:<tag>`

   For example,

   ```
   podman push iad.ocir.io/ocuocictrng24/iad-aiops-lab13-ocir-1/demo_imagescan_user16:1.0
   ```

Verify If the Image Has Been Pushed

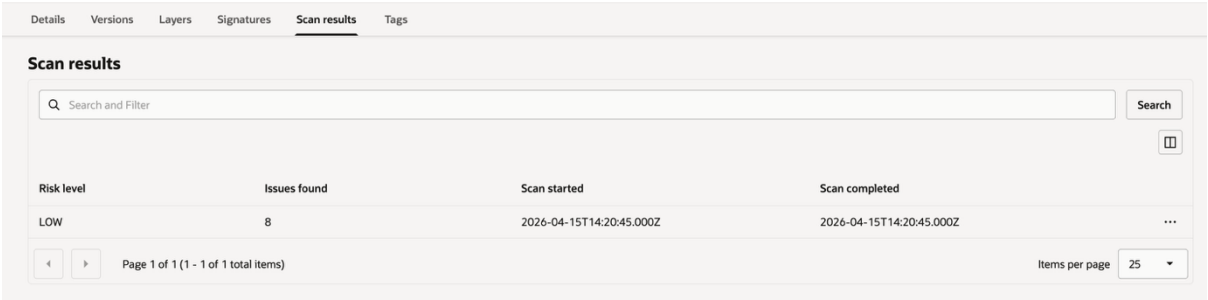
Verify the image was successfully pushed to the OCIR repository.

1. Go back to the OCIR Service page and select your `<assigned compartment>` from List scope on the left menu.
2. You'll see the private repository `iad-aiops-lab13-ocir-1/demo_imagescan_<userID>` that you created.

3. Click the name of the repository that contains the image you just pushed. You'll see:
   - An image with the tag `1.0`.
4. Click on the Image link. You will see various tags like **details, versions, layers and Scan Results** and so on
5. Click the **Scan Results** tab. This will show you the following info:

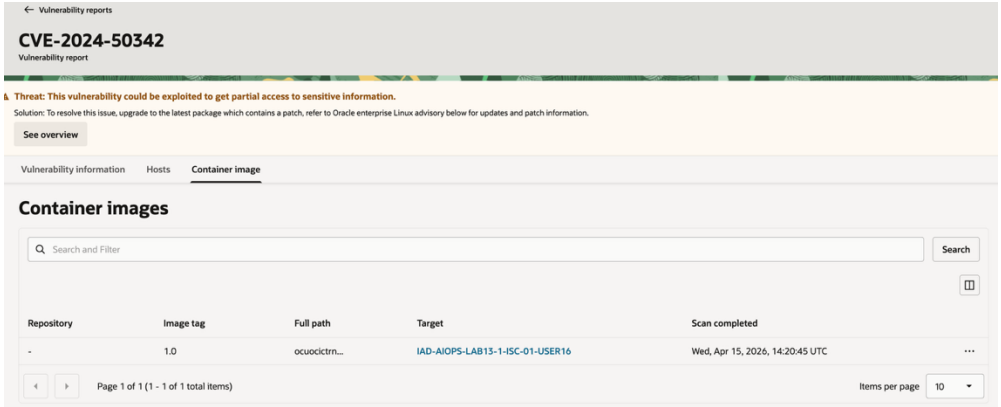
   - Risk level
   - Issues found
   - Scan started
   - Scan completed

   Note: It will take some time before the scan results show up.



6. Click the three dots on the right to open the Actions menu. Select **View Details** to see the issues in more detail with risk level associated with each of them and their descriptions.

View Vulnerability Reports

1. Open the navigation menu and click **Identity & Security**. Under **Scanning**, click **Vulnerability Reports**.

2. From the left menu, under **Scanning**, and select **Vulnerability Reports**.
3. Select your `<assigned compartment>` from Applied filters.
4. Click the **Risk level** header to sort by risk level.
5. To view a description of a specific vulnerability, click **Show** in the CVE description column.
6. To view details about a specific vulnerability, click a report's **CVE ID**. This will result in your viewing a vulnerabilities report. A vulnerabilities report includes details about the **affected resources** and **CVEs**.
7. On the Vulnerabilities report page, in the left menu, under **Resources**, select **Container Images** to view a list of Container images that are affected by the selected vulnerability.